ASSIGNMENT 1 Solutions: Multiple Linear Regression

First order Model with Interaction Term (Quantitative Variables)

Deadline: November 3, 2024, by 11:59 pm. Submit to Gradescope.ca

Problem 1. (From Exercise 1) The amount of water used by the production facilities of a plant varies. Observations on water usage and other, possibility related, variables were collected for 250 months. The data are given in **water.csv file**. The explanatory variables are

TEMP= average monthly temperature (degree celsius)

PROD=amount of production (in hundreds of cubic)

DAYS=number of operationing day in the month (days)

HOUR=number of hours shut down for maintenance (hours)

The response variable is USAGE=monthly water usage (gallons/minute)

a. Fit the model containing all four independent variables. What is the estimated multiple regression equation?

```
waterdata = read.csv("https://raw.githubusercontent.com/DanikaLipman/DATA603/main/water.csv")
Advertising=read.table("https://raw.githubusercontent.com/DanikaLipman/DATA603/main/Advertising.txt",header = TRUE,  sep
="\t" )
fullmodel=lm(USAGE~.,data=waterdata)
summary(fullmodel)
```

```
##
## Call:
## lm(formula = USAGE ~ ., data = waterdata)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.4030 -1.1433  0.0473  1.1677  5.3999
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.891627   1.028794   5.727  3.0e-08 ***
## PROD         0.040207   0.001629  24.681  < 2e-16 ***
## TEMP         0.168673   0.008209  20.546  < 2e-16 ***
## HOUR        -0.070990   0.016992  -4.178  4.1e-05 ***
## DAYS        -0.021623   0.032183  -0.672    0.502
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.768 on 244 degrees of freedom
## Multiple R-squared:  0.8885, Adjusted R-squared:  0.8867
## F-statistic:   486 on 4 and 244 DF,  p-value: < 2.2e-16
```

We fit the model using the command lm in R. See R code above. The estimated multiple regression equation is $us\hat{a}ge = 5.891627 + 0.040207PROD + 0.168673TEMP - 0.070990HOUR - 0.021623DAYS$.

b. Test the hypothesis for the full model i.e the test of overall significance. Use significance level 0.05. Ensure that you include the hypotheses, p-value, and conclusion.

$$H_0 : \beta_i = 0 \; \forall i \;\; vs \;\; H_a : \beta_i \neq 0 \text{ forat least one i}$$

With a p-value < 2.2e-16 <0.05 so we reject H_0 at α = 0.05. Therefore, at least one of the predictors must be related to the response water usage.

c. Would you suggest the model in part b for predictive purposes? Which model or set of models would you suggest for predictive purposes? Hint: Use Individual Coefficients Test (t-test) to find the best model. Make sure you provide your hypotheses, p-value(s), and final model.)

The t-test tests the hypothesis $H_0\beta_i = 0$versus $H_a : \beta_i \neq 0$ for a specific i. From the summary above we see that the p-value for DAYS is <.05, so we will reject the null hypothesis that DAYS is significant and fit the model without DAYS.

```
reducedmodel=lm(USAGE~PROD+TEMP+HOUR,data=waterdata)
summary(reducedmodel)
```

```
##
## Call:
## lm(formula = USAGE ~ PROD + TEMP + HOUR, data = waterdata)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.5066 -1.1356  0.0469  1.1519  5.3750
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.307511   0.549483   9.659  < 2e-16 ***
## PROD         0.040115   0.001621  24.741  < 2e-16 ***
## TEMP         0.169188   0.008164  20.723  < 2e-16 ***
## HOUR        -0.070769   0.016970  -4.170 4.23e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.766 on 245 degrees of freedom
## Multiple R-squared:  0.8883, Adjusted R-squared:  0.8869
## F-statistic: 649.3 on 3 and 245 DF,  p-value: < 2.2e-16
```

We now see that all predictors are significant, so we will use the model:
$us\hat{a}ge = 5.3075 + 0.040115PROD + 0.169188TEMP - 0.070769HOUR.$

d. Use Partial F test to confirm that the independent variable (removed from part c) should be out of the model at significance level 0.05. Ensure that you include the hypotheses, p-value, and conclusion.

$H_0 : \beta_4 = 0$ versus $H_a : \beta_4 \neq 0$ in the model $us\hat{a}ge = \beta_0 + \beta_1 PROD + \beta_2 TEMP + \beta_3 HOUR + \beta_4 DAYS.$

```
anova(reducedmodel,fullmodel)
```

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	245	764.4699	NA	NA	NA	NA
2	244	763.0582	1	1.411739	0.4514261	0.5022941

2 rows

With a p-value of .5023 > 0.05 we fail to reject the null hypothesis and accept our reduced model.

e. Obtain a 95% confidence interval of regression coefficient for TEMP from the model in part c. Give an interpretation.

```
confint(reducedmodel)
```

```
##                   2.5 %      97.5 %
## (Intercept)  4.22519744  6.38982411
## PROD         0.03692098  0.04330837
## TEMP         0.15310634  0.18526907
## HOUR        -0.10419445 -0.03734272
```

The 95% confidence interval for regression coefficient for TEMP is (0.15310634,0.18526907). This means that we conclude with 95% confidence that the monthly water usage increases by 0.153106434 (gallons/minute) to 0.18526907 (gallons/minute) for every 1 degree Celsius increase in average temperature.

f. Use the method of Model Fit to calculate R^2_{adj} and RSE to compare the full model in part a and the model in part c. Which model or set of models would you suggest for predictive purpose? For the final model, give an interpretation of R^2_{adj} and RSE.

For the full model R^2_{adj} = 0.8867 and RSE = 1.768. For the reduced model R^2_{adj} = 0.8869 and RSE = 1.766.The R^2_{adj} is higher for and the RSE lower, for the reduced model. This in combination with the predictor significance confirms the reduced model is better.

The R^2_{adj} means that 88.69% of the variation of the water usage is explained by the model. The RSE = 1.766 means that the standard deviation of the unexplained variance by the model is 1.766 (gallons/minute).

g. Build an interaction model to fit the multiple regression model from the model in part f. From the output, which model would you recommend for predictive purposes? Be sure to explain your process.

```
interacmodel1=lm(USAGE~(PROD+TEMP+HOUR)^2,data=waterdata)
summary(interacmodel1)
```

```
##
## Call:
## lm(formula = USAGE ~ (PROD + TEMP + HOUR)^2, data = waterdata)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.1941 -0.3165 -0.0502  0.2755  7.0985
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.294e+01  7.113e-01  18.193   <2e-16 ***
## PROD        -3.642e-03  2.565e-03  -1.420    0.157
## TEMP        -2.389e-02  2.129e-02  -1.122    0.263
## HOUR        -2.340e-01  2.512e-02  -9.316   <2e-16 ***
## PROD:TEMP    1.189e-03  6.932e-05  17.154   <2e-16 ***
## PROD:HOUR    7.767e-04  7.820e-05   9.933   <2e-16 ***
## TEMP:HOUR    7.600e-04  7.683e-04   0.989    0.324
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9867 on 242 degrees of freedom
## Multiple R-squared:  0.9656, Adjusted R-squared:  0.9647
## F-statistic:  1131 on 6 and 242 DF,  p-value: < 2.2e-16
```

```
interacmodel2=lm(USAGE~PROD+TEMP+HOUR+PROD*TEMP+PROD*HOUR,data=waterdata)
summary(interacmodel2)$coef
```

```
##                  Estimate   Std. Error     t value     Pr(>|t|)
## (Intercept) 12.4257346600 4.838861e-01  25.6790491 3.511910e-71
## PROD        -0.0025288273 2.304825e-03  -1.0971887 2.736450e-01
## TEMP        -0.0047367734 8.858560e-03  -0.5347115 5.933383e-01
## HOUR        -0.2150726648 1.624154e-02 -13.2421388 1.692880e-30
## PROD:TEMP    0.0011417022 5.008616e-05  22.7947652 2.774522e-62
## PROD:HOUR    0.0007873227 7.745417e-05  10.1650127 1.865299e-20
```

We fit the following models in order to include potential interactions: i) we first include all possible interactions with variables involved in the model obtained in part f. Then ii) we remove individually interactions that are not significant. The R code to implement this procedure is shown above.

After fitting the model with all interactions (interacmodel1), interaction TEMP x HOUR was not significant and we removed it from model interacmodel2. Model interacmodel2 has R^2_{adj} = 0.9647 (higher than reducedmodel) and RSE = 0.9866 (lower than the reducedmodel). Hence we will recommend model interacmodel2 with interactions PROD x TEMP and PROD x HOUR. because of it's R^2_{adj}, RSE, and significant predictors.

The final model is:

$$us\hat{a}ge = 12.43 - 0.0025PROD - 0.0047TEMP - 0.2151HOUR + 0.0011PROD * TEMP + 0.0007PROD * HOUR.$$

Problem 2. A collector of antique grandfather clocks sold at auction believes that the price received for the clocks depends on both the age of the clocks and the number of bidders at the auction. Thus, (s)he hypothesizes the first-order model

$$\begin{aligned} y &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon \\ where & \\ y &= \text{Auction price (dollars)} \\ X_1 &= \text{Age of clock (years)} \\ X_2 &= \text{Number of bidders} \end{aligned}$$

A sample of 32 auction prices of grandfather clocks, along with their age and the number of bidders, is given in data file **GFCLOCKS.CSV**

a. Use the method of least squares to estimate the unknown parameters β_0, β_1, β_2 of the model.

```
clock = read.csv("https://raw.githubusercontent.com/DanikaLipman/DATA603/main/GFCLOCKS.csv")
fullmodel=lm(PRICE~AGE+NUMBIDS,data=clock)
summary(fullmodel)
```

```
##
## Call:
## lm(formula = PRICE ~ AGE + NUMBIDS, data = clock)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -206.49 -117.34   16.66  102.55  213.50
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1338.9513   173.8095  -7.704 1.71e-08 ***
## AGE            12.7406     0.9047  14.082 1.69e-14 ***
## NUMBIDS        85.9530     8.7285   9.847 9.34e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 133.5 on 29 degrees of freedom
## Multiple R-squared:  0.8923, Adjusted R-squared:  0.8849
## F-statistic: 120.2 on 2 and 29 DF,  p-value: 9.216e-15
```

$\hat{\beta}_0 = -1338.9513, \hat{\beta}_1 = 12.7406 \; and \; \hat{\beta}_2 = 85.9530.$

b. Find the value of SSE that is minimized by the least squares method.

```
anova(lm(PRICE~1,data=clock),fullmodel)
```

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	31	4799789.5	NA	NA	NA	NA
2	29	516726.5	2	4283063	120.1882	9.216359e-15

2 rows

SSE = 516727

c. Estimate s, the standard deviation of the model, and interpret the result.

From the summary function summary(fullmodel), the output reports RSE = 133.5. This value shows the standard deviation of the unexplained variance in Auction price. It shows how far off using the model is from actual value Y.

d. Find and interpret the adjusted coefficient of determination, R^2_{Adj}.

The adjusted R^2 is 0.8849. The percent of variation in auction price that can be explained by using this model is 88.49 %. The rest (11.51%) can be explained by other predictors.

e. Construct the ANOVA table for the model and test the global F-test of the model at the α = 0.05 level of significance. Be sure to state your hypotheses, p-values, and conclusion.

The ANOVA Table

Source of Variation	Df	Sum of squares	Mean squares	F-statistics
Regression	2	4283063	2141532	120.19
Residual	29	516727	17818.17	
Total	31	4799790		

We test $Ho : \beta_1 = \beta_2 = 0$ versus H_a : at least one $\beta_i \neq 0$. Fcal= 120.2 on 2 and 29 DF, with p-value: 9.216e-15 <0.05 so we reject H_o at α = 0.05. Therefore, at least one of the predictors must be related to the Auction price.

f. Test the hypothesis that the mean auction price of a clock changes as the number of bidders increases when age is held constant (i.e., when $\beta_2 \neq 0$). (Use α = 0.05) Be sure to state your hypotheses, p-values, and conclusion.

The test is $H_0 : \beta_2 = 0$ versus $H_a : \beta_2 \neq= 0$. Tcal = 49.847 with p-value= 9.34e-11<0.05, confirming that we reject the null hypothesis and the predictor NUMBIDS should clearly be added into the model at α = 0.05.

g. Find a 95% confidence interval for β_1 and interpret the result.

```
confint(fullmodel)
```

```
##                   2.5 %     97.5 %
## (Intercept) -1694.43162 -983.47106
## AGE            10.89017   14.59098
## NUMBIDS        68.10115  103.80482
```

From the output obtained from the command confint(fullmodel), a 95% confidence Interval for AGE is (10.89017, 14.59098) which means that we can conclude with 95% confidence that the auction price increases 10.89017 dollars to 14.59098 dollars for every 1 year.

h. Test the interaction term between the 2 variables at $\alpha = .05$. What model would you suggest to use for predicting y? Explain.

```
interacmodel=lm(PRICE~AGE+NUMBIDS+AGE*NUMBIDS,data=clock)
summary(interacmodel)
```

```
##
## Call:
## lm(formula = PRICE ~ AGE + NUMBIDS + AGE * NUMBIDS, data = clock)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.995  -70.431    2.069   47.880  202.259
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 320.4580   295.1413   1.086  0.28684
## AGE           0.8781     2.0322   0.432  0.66896
## NUMBIDS     -93.2648    29.8916  -3.120  0.00416 **
## AGE:NUMBIDS   1.2978     0.2123   6.112 1.35e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 88.91 on 28 degrees of freedom
## Multiple R-squared:  0.9539, Adjusted R-squared:  0.9489
## F-statistic:   193 on 3 and 28 DF,  p-value: < 2.2e-16
```

To test if we should include the interaction we are testing $H_0 : \beta_{interaction} = 0$ versus $H_a : \beta_{interaction} \neq 0$. Fro the R output we have a p-value that is 1.35x10^-6, which is much smaller than alpha so we can reject the null hypothesis and conclude that we should keep the interaction in the model. The model also has a lower RSE of 88.91, and higher R^2_{adj} of 0.9489, meaning it has better prediction performance.

Problem 3. **Cooling method for gas turbines.** Refer to the Journal of Engineering for Gas Turbines and Power (January 2005) study of a high pressure inlet fogging method for a gas turbine engine. The heat rate (kilojoules per kilowatt per hour) was measured for each in a sample of 67 gas turbines augmented with high pressure inlet fogging. In addition, several other variables were measured, including cycle speed (revolutions per minute), inlet temperature (degree celsius), exhaust gas temperature (degree Celsius), cycle pressure ratio, and air mass flow rate (kilograms persecond). The data are saved in the **TURBINE.CSV** file.

a. Write a first-order model for heat rate (y) as a function of speed, inlet temperature, exhaust temperature, cycle pressure ratio, and air flow rate.

```
turbine = read.csv("https://raw.githubusercontent.com/DanikaLipman/DATA603/main/TURBINE.csv")
fullmodel=lm(HEATRATE~RPM+INLET.TEMP+EXH.TEMP+CPRATIO+AIRFLOW,data=turbine)
summary(fullmodel)
```

```
##
## Call:
## lm(formula = HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + CPRATIO +
##     AIRFLOW, data = turbine)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1007.0  -290.9  -105.8   240.8  1414.0
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.361e+04  8.700e+02  15.649  < 2e-16 ***
## RPM          8.879e-02  1.391e-02   6.382 2.64e-08 ***
## INLET.TEMP  -9.201e+00  1.499e+00  -6.137 6.86e-08 ***
## EXH.TEMP     1.439e+01  3.461e+00   4.159 0.000102 ***
## CPRATIO      3.519e-01  2.956e+01   0.012 0.990539
## AIRFLOW     -8.480e-01  4.421e-01  -1.918 0.059800 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 458.8 on 61 degrees of freedom
## Multiple R-squared:  0.9235, Adjusted R-squared:  0.9172
## F-statistic: 147.3 on 5 and 61 DF,  p-value: < 2.2e-16
```

From the command line summary(fullmodel) (see R code above) after using the least square method, we can write the first-order model for heart rate as

$$\widehat{HEATRATE} = 13610 + 0.08879RPM - 9.201INLET.TEMP + 14.39EXH.TEMP + 0.3519CPRATIO - 0.848AIRFLOW$$

b. Test the overall significance of the model using $\alpha = 0.01$. Be sure to state your hypotheses, p-values, and conclusion.

We would like to test $H_0 : \beta_i = 0 \; \forall i$ versus $H_a : at\ least\ one\ \beta_i \neq 0$. From the output of summary(fullmodel), Fcal= 147.3 on 5 and 61 DF, p-value: <2.2e-16 <0.01 so we reject H_o at α = 0.05. Therefore, at least one of the predictors must be related to heatrate (Y).

c. Fit the best additive model to the data using the method of least squares. Test significance of predictors at $\alpha = 0.06$. Be sure to state your hypotheses, p-values, and conclusion.

For each predictor we test the hypothesis we would like to test $H_0: \beta_i = 0$ versus $H_a: \beta_i \neq 0$. I start by removing CPRATIO because it is not significant with a p-value of 0.990539.

```
reducemodel=lm(HEATRATE~RPM+INLET.TEMP+EXH.TEMP+AIRFLOW,data=turbine)
summary(reducemodel)
```

```
##
## Call:
## lm(formula = HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + AIRFLOW,
##     data = turbine)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1007.7  -290.5  -106.0   240.1  1414.8
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.362e+04  8.133e+02  16.744  < 2e-16 ***
## RPM          8.882e-02  1.344e-02   6.608 1.02e-08 ***
## INLET.TEMP  -9.186e+00  7.704e-01 -11.923  < 2e-16 ***
## EXH.TEMP     1.436e+01  2.260e+00   6.356 2.76e-08 ***
## AIRFLOW     -8.475e-01  4.370e-01  -1.939    0.057 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 455.1 on 62 degrees of freedom
## Multiple R-squared:  0.9235, Adjusted R-squared:  0.9186
## F-statistic: 187.1 on 4 and 62 DF,  p-value: < 2.2e-16
```

This model is better as all predictors are significant our our RSE is lower at 455.1, and the adjusted R squared is higher at 0.9186. Thus the best model is $\widehat{HEATRATE} = 13618 + 0.088823RPM - 9.1856INLET.TEMP + 14.36EXH.TEMP - 0.848AIRFLOW$

d. Test all possible interaction terms for the best model in part (c) at $\alpha = .06$. What is the final model would you suggest to use for predicting y? Explain.

Start with the full interaction model:

```
interacmodel1=lm(HEATRATE~(RPM+INLET.TEMP+EXH.TEMP+AIRFLOW)^2,data=turbine)
summary(interacmodel1)
```

```
##
## Call:
## lm(formula = HEATRATE ~ (RPM + INLET.TEMP + EXH.TEMP + AIRFLOW)^2,
##     data = turbine)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -779.7 -211.0  -40.7  177.2 1370.3
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          2.650e+04  8.891e+03   2.981 0.004247 **
## RPM                  7.037e-02  1.485e-01   0.474 0.637512
## INLET.TEMP          -2.366e+01  7.364e+00  -3.213 0.002180 **
## EXH.TEMP            -4.555e+00  1.795e+01  -0.254 0.800610
## AIRFLOW              1.021e+01  6.279e+00   1.627 0.109455
## RPM:INLET.TEMP      -1.133e-04  8.720e-05  -1.299 0.199266
## RPM:EXH.TEMP         1.656e-04  3.116e-04   0.531 0.597314
## RPM:AIRFLOW         -8.257e-04  4.653e-04  -1.775 0.081414 .
## INLET.TEMP:EXH.TEMP  2.417e-02  1.457e-02   1.659 0.102791
## INLET.TEMP:AIRFLOW   1.418e-02  3.852e-03   3.681 0.000523 ***
## EXH.TEMP:AIRFLOW    -5.049e-02  1.357e-02  -3.720 0.000463 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 394.6 on 56 degrees of freedom
## Multiple R-squared:  0.9481, Adjusted R-squared:  0.9388
## F-statistic: 102.3 on 10 and 56 DF,  p-value: < 2.2e-16
```

Remove the least significant interactions:

```
interacmodel2=lm(HEATRATE~RPM+INLET.TEMP+EXH.TEMP+AIRFLOW+INLET.TEMP*AIRFLOW+EXH.TEMP*AIRFLOW+RPM*AIRFLOW,data=turbine)
summary(interacmodel2)
```

```
##
## Call:
## lm(formula = HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + AIRFLOW +
##     INLET.TEMP * AIRFLOW + EXH.TEMP * AIRFLOW + RPM * AIRFLOW,
##     data = turbine)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -791.20 -196.73  -41.25  187.07 1272.39
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.331e+04  1.006e+03  13.227  < 2e-16 ***
## RPM                 3.915e-02  1.631e-02   2.400 0.019555 *
## INLET.TEMP         -1.264e+01  1.087e+00 -11.633  < 2e-16 ***
## EXH.TEMP            2.402e+01  2.923e+00   8.219 2.33e-11 ***
## AIRFLOW             2.876e+00  3.627e+00   0.793 0.431060
## INLET.TEMP:AIRFLOW  1.660e-02  3.623e-03   4.582 2.43e-05 ***
## EXH.TEMP:AIRFLOW   -4.278e-02  1.082e-02  -3.954 0.000208 ***
## RPM:AIRFLOW        -6.219e-04  4.347e-04  -1.431 0.157812
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 397.9 on 59 degrees of freedom
## Multiple R-squared:  0.9444, Adjusted R-squared:  0.9378
## F-statistic: 143.1 on 7 and 59 DF,  p-value: < 2.2e-16
```

```
interacmodel3=lm(HEATRATE~RPM+INLET.TEMP+EXH.TEMP+AIRFLOW+INLET.TEMP*AIRFLOW+EXH.TEMP*AIRFLOW,data=turbine)
summary(interacmodel3)
```

```
##
## Call:
## lm(formula = HEATRATE ~ RPM + INLET.TEMP + EXH.TEMP + AIRFLOW +
##     INLET.TEMP * AIRFLOW + EXH.TEMP * AIRFLOW, data = turbine)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -787.68 -189.26  -22.34  145.15 1307.53
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.360e+04  9.930e+02  13.699  < 2e-16 ***
## RPM                 4.578e-02  1.577e-02   2.902 0.005174 **
## INLET.TEMP         -1.280e+01  1.090e+00 -11.741  < 2e-16 ***
## EXH.TEMP            2.327e+01  2.901e+00   8.024 4.46e-11 ***
## AIRFLOW             1.347e+00  3.496e+00   0.385 0.701414
## INLET.TEMP:AIRFLOW  1.613e-02  3.640e-03   4.432 4.03e-05 ***
## EXH.TEMP:AIRFLOW   -4.150e-02  1.087e-02  -3.816 0.000323 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 401.4 on 60 degrees of freedom
## Multiple R-squared:  0.9424, Adjusted R-squared:  0.9367
## F-statistic: 163.7 on 6 and 60 DF,  p-value: < 2.2e-16
```

The final model we will use is:

$$HEATRATE = 13603.309 + 0.046RPM - 12.799INLET.TEMP + 23.274EXH.TEMP + 1.347AIRFLOW + 0.016INLET.TEMP * AIR$$

WE choose this model because even though the RSE is a bit higher and the adjusted Rsquare is a bit lower than the full interaction model, we have significance of all terms, so we avoid overfitting. We prefer this model over the additive model because the RSE is lower and adjusted Rsquare is higher.

e. Give practical interpretations of the β_i estimates.

Note that the final model has two significant interactions. Hence, we will not interpret directly the main effects for variables involved in the interaction term.

1. $\hat{\beta}_{RPM} = 0.0457$ means that for a given amount of other predictors (are held constant), an increase of 1 revolution per minute of the cycle speed leads to an increase in the heat rate by 0.04578 revolutions per minute.
2. The effect of INLET.TEMP is 12.80+0.016AIRFLOW, means that for a given amount of EXH.TEMPand RPM (are held constant), an increase of 1 degree Celsius of the inlet temperature leads to an increase inthe heat rate by -12.80+0.016AIRFLOW revolutions per minute.
3. The effect of EXH.TEMP is 23.27-0.041AIRFLOW, means that for a given amount of INLET.TEMP and RPM (are held constant), an increase of 1 degree Celsius of the exhaust gas temperature leads to an increase in the heat rate by 23.27-0.041AIRFLOW revolutions per minute.

4. The effect of AIRFLOW is 1.34+0.016INLET.TEMP-0.041EXH.TEMP, means that for a given amount of RPM (is held constant), an increase of 1 kilogram per second of air mass flow rate leads to an increase in the heat rate by 1.34+0.016INLET.TEMP-0.041EXH.TEMP revolutions per minute.

f. Find RSE, s from the model in part (d)

RSE=s=401.4

g. Find the adjusted-R2 value from the model in part (d) and interpret it.

The adjusted R^2 is $R^2_{Adj} = 0.9367$ implies that 93.67% of the variation in the heart rate is explained by this model.

h. Predict a heat rate (y) when a cycle of speed = 273,145 revolutions per minute, inlet temperature= 1240 degree celsius, exhaust temperature=920 degree celsius, cycle pressure ratio=10 kilograms persecond, and air flow rate=25 kilograms persecond.

```
newdata = data.frame(RPM=273145, INLET.TEMP=1240,EXH.TEMP=920,AIRFLOW=25)
predict(interacmodel3,newdata, interval="predict")
```

```
##        fit      lwr     upr
## 1 31227.97 24067.74 38388.2
```

From the R command predict (see R code above), with 95% confidence , the heat rate (Y) is between 24067.74 revolutions per minute to 38388.2 revolutions per minute when a cycle of speed = 273,145 revolutions per minute, inlet temperature= 1240 degree Celsius, exhaust temperature=920 degree Celsius, cycle pressure ratio=10 kilograms per second, and air flow rate=25 kilograms per second.